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NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of NITTE (Deemed to be University)
Sixth Semester B.Tech (CBCS) Degree Examinations

May 2025

IS4221-1 – ADHOC WIRELESS NETWORKS

(For IS)

Duration: 3 Hours

Max. Marks: 100

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.

PART - A: MULTIPLE CHOICE QUESTIONS**20 Marks**

1. What is the primary medium access technique used in IEEE 802.11?
A) FDMA B) CSMA/CA
C) TDMA D) CDMA
2. Which of the following is NOT a common application of ad hoc networks?
A) Military communications B) Disaster recovery
C) Fixed broadband connectivity D) Collaborative computing
3. Which of the following is a key design goal of MAC protocols in ad hoc networks?
A) Ensuring fixed data rates B) Providing synchronization
C) Handling hidden and exposed terminal problems D) Allowing packet prioritization only
4. In the MARCH protocol, what does "reduced handshake" imply?
A) Faster transmission B) Fewer control packets
C) Simplified protocol implementation D) Lower energy consumption
5. What type of protocol is HRMA classified as?
A) Contention-based with scheduling B) Contention-based with reservation
C) Hybrid protocol D) Pure contention-based
6. Which type of networks benefit most from DPS protocols?
A) Wired networks B) Cellular networks
C) Ad hoc networks D) Satellite networks
7. Which protocol uses multi-hop coordination for scheduling?
A) MACA B) DPS
C) HRMA D) MACAW
8. Which of the following does WRP use to ensure loop-free routing?
A) Sequence numbers B) Path vectors
C) Update and acknowledgment messages D) Link costs
9. Which protocol is considered more efficient in handling frequent topology changes in ad hoc networks?
A) AODV B) OLSR
C) TORA D) DSDV
10. What is the primary goal of FORP?
A) Minimize routing overhead B) Establish stable routes for continuous data flows
C) Reduce energy consumption D) Support multicast routing
11. What is the key feature of ZRP?
A) Scalability in large networks B) Hierarchical clustering
C) Dividing the network into zones D) Dynamic bandwidth allocation

12. **Why is it called the Fisheye State Routing Protocol?**
 A) It uses a fish-shaped topology B) It provides better routing accuracy for nearby nodes
 C) It avoids long-distance routing D) It is optimized for underwater networks
13. **What is the primary challenge in designing transport layer protocols for ad hoc wireless networks?**
 A) High energy consumption B) Mobility-induced link failures
 C) Lack of encryption mechanisms D) Overhead in centralized management
14. **Which TCP variant uses a snooping mechanism to improve performance in ad hoc networks?**
 A) TCP-F B) TCP-BuS
 C) ATCP D) TCP with ELFN
15. **Which of the following is NOT a core network security requirement?**
 A) Authentication B) Confidentiality
 C) Bandwidth optimization D) Integrity
16. **What is the main purpose of key management in network security?**
 A) Routing optimization B) Ensuring secure key distribution and storage
 C) Increasing bandwidth utilization D) Enhancing mobility
17. **What cryptographic mechanism does SEAD (Secure Efficient Ad hoc Distance Vector Routing Protocol) use?**
 A) Public-key cryptography B) One-way hash functions
 C) Symmetric key encryption D) Digital signatures
18. **What is the primary goal of a blackhole attack?**
 A) Eavesdropping on communications B) Dropping all packets passing through a malicious node
 C) Delaying packet transmission D) Increasing routing table size
19. **Which security requirement ensures that data sent by the sender can only be understood by the intended receiver?**
 A) Integrity B) Confidentiality
 C) Non-repudiation D) Availability
20. **What is the primary characteristic of external attacks in ad hoc wireless networks?**
 A) They are launched by unauthorized nodes outside the network. B) They involve snooping on transmitted data.
 C) They exploit routing table vulnerabilities. D) They inject malicious data into the network.

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I		Marks	BT*	CO*	PO*
1.	a) Explain different types of Inter-Frame Spacing.	8	L*2	1	1,2
	b) List and describe the issues in designing MAC protocol for ad hoc wireless networks.	8	L2	2	1,2
2.	a) Explain Floor acquisition multiple access protocol.	8	L2	1	1,2
	b) Explain any 4 issues in Ad Hoc Wireless Networks.	8	L2	1	1,2
3.	a) List out Design Goals of MAC Protocol For Ad Hoc Wireless Networks.	8	L2	2	1,2
	b) Explain Distributed Packet Reservation Multiple Access (DPRMA) Protocol.	8	L2	2	1,2
Unit – II					
4.	a) Outline the DSDV routing protocol.	8	L2	3	1,2
	b) List out the advantages and disadvantages of TCP-BuS.	8	L2	4	1,2
5.	a) Explain DSR protocol.	8	L2	3	1,2

b)	List out the advantages and disadvantages of ZRP.	8	L2	4	1,2
6. a)	Explain CEDAR protocol.	8	L2	3	1,2
b)	Explain the Issues in Designing a Transport Layer Protocol For Ad Hoc Wireless Networks.	8	L2	4	1,2
Unit – III					
7. a)	Explain Security Aware Ad Hoc Routing (SAR) Protocol.	8	L2	5	1,2
b)	Explain the challenges in security provisioning.	8	L2	5	1,2
8. a)	Explain the concept of Byzantine attack and Resource consumption attack.	8	L2	5	1,2
b)	Explain the Issues and challenges in providing QoS in Adhoc Wireless Networks.	8	L2	5	1,2

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
